

PYTHON FOR PLATFORM ENGINEERS · ISSUE 03 OF 04

Python validation scripts in CI.

Catch the mistake in the PR, not in the cluster.

03

AR

Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

Good candidates

- ☒ YAML and JSON parse cleanly
- ☒ Required keys and labels present
- ☒ No :latest image tags
- ☒ Resource requests set
- ☒ No plaintext secrets in config

Validate manifests

PYTHON

```
import pathlib, sys, yaml

errors = []
for f in pathlib.Path("k8s").rglob("*.yaml"):
    for doc in yaml.safe_load_all(f.read_text()):
        if not doc or doc.get("kind") != "Deployment":
            continue
        spec = doc["spec"]["template"]["spec"]
        for c in spec["containers"]:
            if c["image"].endswith(":latest"):
                errors.append(f"{f}: {c['name']} uses :latest")
            if "resources" not in c:
                errors.append(f"{f}: {c['name']} has no resources")

print("\n".join(errors))
sys.exit(1 if errors else 0)
```



Annotate the PR

```

# GitHub Actions turns this into an inline comment
print(f"::error file={path},line={line}::{message}")

# Azure DevOps equivalent
print(f"##vso[task.logissue type=error;"
      f"sourcepath={path};linenumber={line}]{message}")
```

File and line in the output means the developer fixes it without asking you.

Feedback loop

1 Push

2 Script runs

3 Inline annotations

4 Fix

5 Green check

Seconds, not a code-review round trip.



Script habits

DO

- ✓ `yaml.safe_load`, never `yaml.load`
- ✓ Report every error, not just the first
- ✓ Unit-test with fixture files

AVOID

- ✗ Regex-parsing YAML
- ✗ Silent passes on parse errors
- ✗ Checks that only run on main

KEY TAKEAWAY

Turn team conventions into checks with file-level feedback.

- Small scripts for team-specific rules
- `safe_load` and full error lists
- Annotations instead of review comments

YOUR TURN

Which convention does your team keep repeating in code review?

FOLLOW AKHILESH RANJAN SINGH

Next in the series: #04 Python reports for cost, inventory and incidents

Save · Repost

